Introduction to Large Language Models (LLMs)

Your Journey into AI Engineering

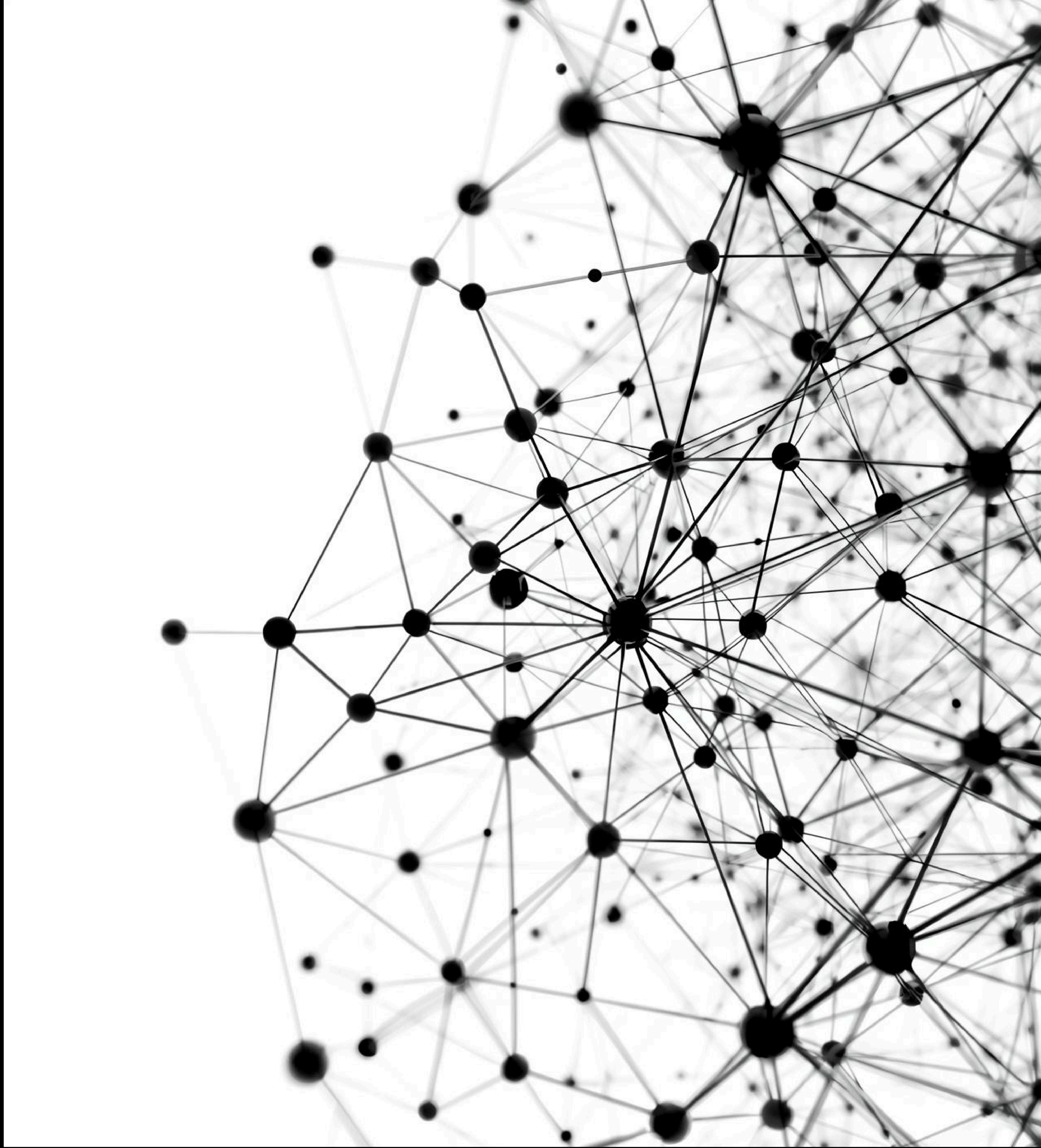
What We'll Cover Today

Learning Objectives for Today

- Understanding what LLMs are
- Running models locally with Ollama
- Navigating the landscape of AI
- Making API calls to GPT
- Mastering system vs user prompts
- Building a website summarizer

What is a Large Language Model?

An AI system trained on massive text data that can understand and generate human-like text. It predicts the next word based on context, using billions of learned parameters.



The Scale of LLMs

TRAINING DATA

Trillions of words sourced from diverse materials such as books, websites, and articles are employed during the training process to develop a comprehensive language understanding.

MODEL SIZE

Ranging from 270 million to an astonishing 1.7 trillion parameters, large language models leverage extensive data to enhance their capability to understand and generate human-like text.

COMPUTE POWER

Thousands of GPUs work tirelessly over weeks or even months to train these models, resulting in systems capable of writing, coding, analyzing, and reasoning effectively.

What is Ollama?

THE TOOL BEHIND LOCAL AI MODELS

A free application that runs AI models on **YOUR computer**, enabling users to harness the power of AI locally without relying on internet connectivity, ensuring privacy and efficiency.

Key Features

CROSS-PLATFORM COMPATIBILITY

Works seamlessly on Windows, Mac, and Linux operating systems without issues.

USER-FRIENDLY INTERFACE

Simple command-line interface makes it easy for users to operate the application.

AUTOMATED MANAGEMENT

Downloads and manages AI models automatically, saving time and effort for users.

OFFLINE FUNCTIONALITY

No internet is needed after the initial download, ensuring uninterrupted access.

LOCAL AI ADVANTAGE

Privacy, free usage, and speed combine for a powerful local AI experience.

Why Ollama is Important

PRIVACY FIRST

Ollama operates entirely on your device, ensuring that your data remains **private** and secure without needing to be shared with external servers.

COST EFFECTIVE

Being a free application, Ollama provides powerful AI capabilities at no cost, making advanced technology accessible to everyone without financial barriers.

Running Your First AI Model

COMMAND TO INITIATE THE MODEL

To begin using Ollama,
enter the command: **ollama run gemma3:270m**.
This command starts Google's Gemma 3 model with 270 million parameters, allowing for efficient AI interactions.

What This Does

THE COMMAND

ollama run initiates the process of starting a specific model, enabling users to utilize the functionalities of the AI model on their computer seamlessly.

GOOGLE'S MODEL

gemma3 refers to Google's Gemma 3 model, a powerful AI model designed to execute various tasks efficiently and effectively within the ollama framework.

PARAMETER VERSION

:270m indicates the 270 million parameter version of the Gemma 3 model, signifying its complexity and capability in delivering advanced AI solutions to users.

Understanding Model Sizes

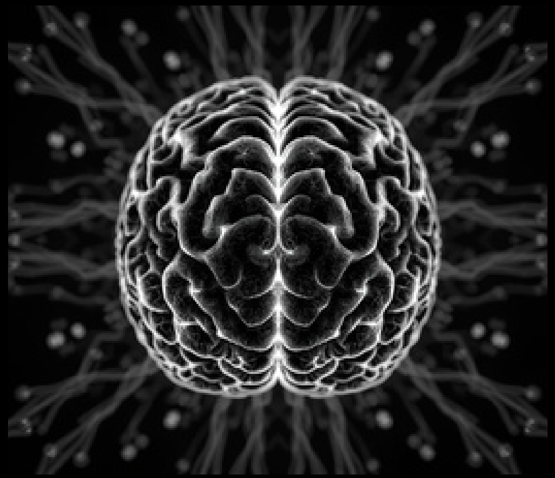
Size	Parameters	RAM Needed	Speed
Tiny	270M-1B	1-2 GB	Very Fast
Small	1-3B	2-4 GB	Fast
Medium	7-8B	8-12 GB	Medium
Large	70B+	40GB+	Slow

Rule of Thumb
~1 GB RAM per 1 billion parameters

Models Available in Ollama

Gemma

Created by Google, ranging from 270M–27B parameters; designed for reliable general-purpose tasks and lightweight deployments.



Llama

Built by Meta, spanning 1B–405B parameters; a strong all-around model family for research and production.

Phi

From Microsoft, around 3.8B parameters; optimized for efficiency and smaller-scale reasoning.

Mistral

Developed by Mistral AI, typically 7B parameters; known for high performance relative to size.

Qwen

Created by Alibaba, ranging from 0.5B–110B parameters; offers strong versatility across tasks and scales.

DeepSeek

From DeepSeek, with models of various sizes; focused on advanced reasoning capabilities.

Full model list available at *ollama.com/library*.

Frontier Models

OPENAI

OpenAI is known for their models GPT-4 and GPT-5, which set the **industry standard** for versatility and performance in various applications across multiple sectors.

ANTHROPIC

Anthropic's model, Claude, is recognized for its **nuanced** and thoughtful approach to AI, focusing on ethical considerations and human-like understanding in interactions with users.

GOOGLE

Google's Gemini model stands out for being **multimodal** and integrated, allowing for seamless transitions between text and visual data, enhancing user experience and functionality.

XAI

xAI has developed Grok, a model that offers **real-time information** capabilities, making it a valuable tool for applications requiring immediate and up-to-date responses in dynamic environments.

Frontier Models Characteristics

MOST POWERFUL MODELS

The **most powerful models** available today are designed to handle complex tasks and deliver exceptional results across various applications, making them essential for businesses and researchers.

HIGH TRAINING COSTS

The training costs for these models exceed **$100M**, reflecting the extensive resources and expertise required to develop and maintain such advanced artificial intelligence systems.

PAID ACCESS

These models are primarily accessed through **paid APIs** or subscriptions, ensuring that users have the necessary resources and support to effectively utilize these cutting-edge technologies.

INDUSTRY IMPACT

With their superior performance and accessibility, these models significantly impact industries, driving innovation and enhancing productivity in a competitive landscape.

Open Source Models

LLAMA

Most popular open source model, widely used in various applications and projects.

MISTRAL

Known for its excellent efficiency, delivering high performance with lower resource consumption.

QWEN

An underrated powerhouse, Qwen demonstrates strong capabilities in diverse AI tasks.

GEMMA

A small but capable model, Gemma offers effective solutions for specific needs.

DEEPSEEK

Near-GPT performance at a low cost, making it an attractive alternative for users.

Open Source Models

FREE

Open source models are available at no cost to users and developers.

PRIVATE

Users maintain control over their data without third-party interference or monitoring.

CUSTOMIZABLE

These models can be tailored to meet specific user needs and preferences.

NO INTERNET

Functionality can be retained without an active internet connection for operations.

VERSATILE

Open source models can adapt to various applications and use cases effectively.

Closed vs Open Source Comparison

Aspect	Closed-Source	Open-Source
Cost	Pay per use	Free
Privacy	Data sent to servers	Stays on your machine
Capability	Generally higher	Rapidly Improving
Customization	Limited	Full Control
Access	API/Subscription	Download freely
Internet	Required	Optional

Three Ways to Use LLMs

Exploring Practical Applications for LLMs

Consumer Products

Popular applications like **ChatGPT**, **Claude.ai**, and **Gemini app** offer ease of use and a subscription model, making them accessible for everyday users seeking AI assistance.

Cloud APIs

Tools such as **OpenAI API**, **Anthropic API**, and **Vertex AI** are designed for developers, operating on a pay-per-token basis, facilitating integration into various applications.

Local Inference

Solutions like **Ollama**, **Hugging Face**, and **LM Studio** enable free and private AI processing on local hardware, offering unique advantages for users concerned with data privacy.

The Chat Completions API

```
response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {"role": "user", "content": "Hello!"}
    ]
)
```

The Chat Completions API allows applications to send a conversation to a GPT model. The model processes the conversation and responds in a natural, chat-like manner, similar to a human assistant.

model

Which GPT version to use

messages

List of conversation messages

response

Contains the AI's reply

Understanding Messages Format

```
messages = [
    {"role": "system", "content": "Instructions"},
    {"role": "user", "content": "Question"},
    {"role": "assistant", "content": "Response"}
]
```

A guide to conversation structure

Message Structure

Conversations consist of three types of messages: system, user, and assistant. Each message type plays a vital role in the communication process, presented in a sequential format.

The Three Roles

A structured table outlines the roles involved: the system sets AI behavior, the user provides input, and the assistant delivers responses, ensuring effective interaction.

Visual Example

An example illustrates this format using speech bubbles. The system gives instructions, the user asks questions, and the assistant responds, creating a clear conversational flow.

System Prompt vs User Prompt

- **System Prompt:** Sets the personality and defines task constraints.
- **User Prompt:** The actual question or request for response.
- **Contrast:** System controls HOW; User controls WHAT to respond about.

System Prompt Examples

Illustrating Different Assistant Personas

Helpful Assistant

"You are a helpful assistant." This prompt encourages the AI to provide straightforward and accurate responses, ensuring clarity and precision in its interactions.

Snarky Assistant

"You are a snarky assistant with attitude." This prompt infuses personality into responses, allowing for humorous and sarcastic replies which can add a playful dynamic to the conversation.

Teacher

"You are a patient kindergarten teacher." This prompt fosters a nurturing environment, focusing on educational engagement and encouraging learning through patience and supportive dialogue.

The Power of System Prompts

BE BRIEF

One-sentence answers simplify understanding and encourage concise communication.

BE THOROUGH

Detailed explanations provide clarity and deepen comprehension of complex topics.

SPEAK LIKE A PIRATE

Using playful language adds humor and engagement to the interaction.

RESPOND IN SPANISH

Full Spanish output caters to bilingual users and enhances inclusivity.

FORMAT AS JSON

Structured data allows for easier processing and integration with applications.

KEY INSIGHT

System prompts are your primary control mechanism for effective communication.

Building a Website Summarizer

THE CONCEPT

This process involves fetching content from a URL and generating a summary.

COMPONENTS NEEDED

Essential elements include a web scraper and prompts for summarization tasks.

WEB SCRAPER

The scraper is responsible for retrieving the full content of the webpage.

SYSTEM PROMPT

This defines how the summarization task should be approached and executed.

USER PROMPT

Contains the specific content that needs to be summarized by the system.

The Concept

URL FETCHING

This initial step involves retrieving the webpage content by accessing the URL, allowing subsequent processes to analyze and summarize the extracted information effectively.

SYSTEM PROMPT

The system prompt defines the summarization task, guiding the AI on how to interpret the content and what focus points to prioritize for the generated summary.

WEB SCRAPER

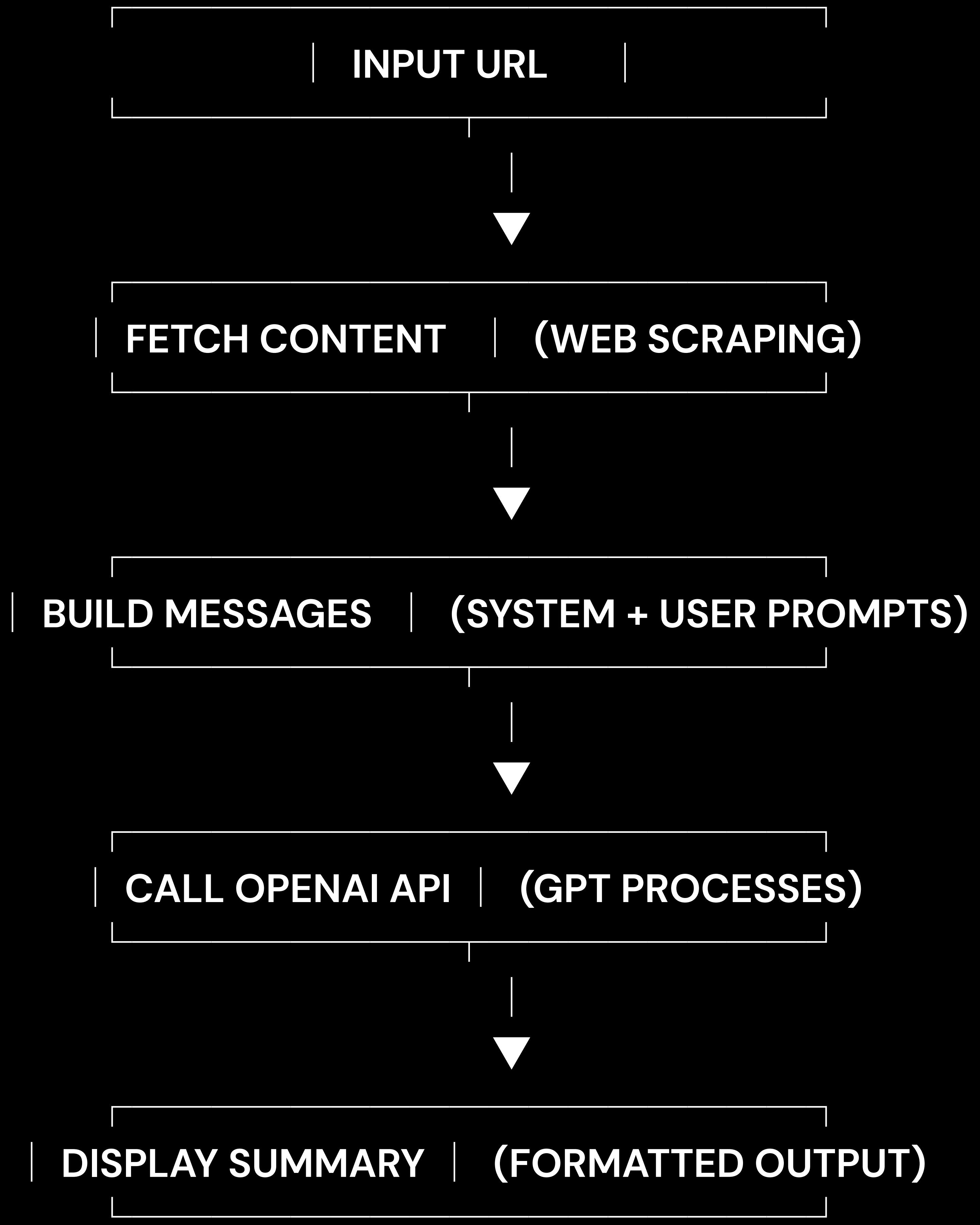
A web scraper is essential for fetching the page content, ensuring all relevant text is captured for the summarization task, enabling an efficient data processing flow.

USER PROMPT

The user prompt contains the actual content that needs summarization, providing context and direction for the AI to produce concise and informative results based on the input.

Summarizer System Prompt

```
system_prompt = """You are an assistant that
analyzes website content and provides concise
summaries.
Guidelines:
  • Focus on main content, ignore navigation
  • Highlight key points
  • Use bullet points
  • Keep under 200 words
  • Respond in markdown format"""
```



Practical Business Applications

01 **Email Management**

Generate subject lines for effective communication.

02 **Content Creation**

Social media posts engage audiences and build presence.

03 **Trend Identification**

Key point extraction aids in strategic planning.

04 **Customer Service**

Ticket categorization helps streamline support processes.

05 **Data Analysis**

Report summaries provide insights for decision making.

Next Steps

Practice Exercises

Try 3 different Ollama models to gain hands-on experience with varying capabilities and functionalities of each model.

System Prompts

Create 5 different system prompts to experiment with and see how they influence the behavior of language models in various scenarios.

Resources

Explore comprehensive documentation from OpenAI, Ollama, and Anthropic to enhance your understanding and application of language models.

Build Tools

Build your own tool, whether it's an email client, code snippet generator, or any other application that utilizes language models effectively.